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The effects of verbal and spatial interference in the encoding and retrieval of spatial and nonspatial texts

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Abstract The paper investigates the specific roles of visual-spatial working memory (VSWM) and verbal working memory (VWM) in encoding and retrieval of information conveyed by spatial and nonspatial texts. In two experiments, a total of 109 undergraduate students—54 in Experiment 1, 55 in Experiment 2—listened to spatial and nonspatial texts while performing a spatial (Experiment 1) or verbal (Experiment 2) concurrent task during either encoding or retrieval. Text memorisation and comprehension were tested by free-recall and sentence-verification tasks. The results show that a concurrent spatial task is detrimental to memory performance for spatial text more than for nonspatial text. In contrast, a concurrent verbal task is equally damaging to memory performance for both spatial and nonspatial texts. Moreover, a spatial task interferes with both encoding and retrieval, in contrast with a verbal task, where the interference effect is active only when the task is performed during encoding. Overall, these findings show the involvement of VSWM in the construction and reactivation of mental models derived from spatial descriptions, and the role played by VWM in construction, but not reactivation, of mental models derived from spatial and nonspatial texts.

Introduction

A particular skill of human beings is their ability to construct spatial mental representations from language or, inversely, to translate such mental representations

into language. These processes require strict cooperation between verbal and spatial systems, and have been the subject of intense study in recent decades (Denis, Logie, Cornoldi, de Vega, & Engelkamp, 2001; Landau & Jackendoff, 1993; Levelt, 1989; Perrig & Kintsch, 1985; Taylor & Tverky, 1992; de Vega, Cocude, Denis, Rodrigo, & Zimmer, 2001). People build mental models to represent significant aspects of their physical world and manipulate them when thinking and planning (Bower & Morrow, 1990). When mental models are derived from spatial descriptions, they maintain the characteristics of the original spatial configuration, as proved by several empirical findings (Baguley & Payne, 2000; de Vega et al., 2001; Morrow, Bower, & Greenspan, 1989; Morrow, Greenspan, & Bower, 1987).

If mental models are spatial in nature, we might expect a temporary spatial memory function to be involved in the encoding and retrieval. The role played by visual-spatial working memory (VSWM) in the encoding of spatial descriptions has been supported by a number of studies (De Beni, Pazzaglia, Gyselinck, & Meneghetti, 2005; Noordzij, van der Lubbe, Neggers, & Postma, 2004; Pazzaglia & Cornoldi, 1999), whose theoretical background was provided by Baddeley's model of working memory (Baddeley, 1986; 2003). In this latter model, working memory (WM) is thought of as a temporary storage and processing system with a central executive plus two slave sub-components: verbal (verbal working memory, VWM) and visuo-spatial (VSWM), dedicated to processing verbal and visual-spatial information, respectively. VWM keeps phonological entries active under the control of an articulated process, while VSWM maintains and manipulates spatial and visual information. A specific involvement of WM's verbal and visual-spatial components in the processing of abstract and spatial sentences was found by Pazzaglia and Cornoldi (1999), whose results showed interference effects for concurrent verbal and spatial tasks on memorisation of abstract and spatial sentences, respectively. When participants listened to lists of abstract and spatial sentences (Brooks, 1967), and concurrently performed either

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verbal or spatial tasks, their recall of abstract sentences was impaired more by the verbal task than the spatial task, but the converse was true for recall of spatial sentences.

De Beni et al. (2005) found similar results when using longer coherent texts instead of lists of sentences. Participants read spatial and nonspatial texts while performing a verbal (articulatory suppression, AS) task—repetition of “ba-be-bi-bo-bu”, or a spatial (spatial tapping, ST) concurrent task—tapping four keys at the corners of a square table. The findings from free-recall and sentence-verification tasks showed that, compared to control conditions, the AS affected the memorisation of both nonspatial and spatial texts, whereas the ST impaired memory performance of only the spatial text. These results show that the verbal and spatial components of WM have distinct roles in the comprehension and memorisation of spatial and nonspatial texts. The verbal component is necessary for processing both spatial and nonspatial texts. In contrast, VSWM has a specific role in spatial text processing alone. Noordzij et al. (2004) also obtained results in line with the idea of involvement of the visual-spatial component of WM in the representation of verbally presented spatial relations. They used a sentence-picture verification task, where participants compared linguistic information in a sentence (e.g. triangle left of circle) with visual-spatial information on a corresponding picture, and concurrently performed an AS or ST task. As expected, participants were significantly slower and less accurate in the sentence-picture verification task when they had to perform the concurrent ST, whereas the AS produced no detrimental effects.

Taken together, the outcomes of the above studies might be interpreted in the light of a multi-level text-processing model. In Kintsch’s model (Kintsch & van Dijk, 1978), text-processing results in three different levels of representation, viz: a surface level, associated with the phonetic or graphemic properties of words; a propositional level, associated with the meaning of individual words and sentences (text base); and a situational level, associated with the situation described in the text (situation or mental model). Mental models preserve the spatial properties of a description, which include relations between landmarks (Perrig & Kintsch, 1985; Taylor & Tversky, 1992). We might suppose VWM to be essential for construction of the first two representational levels (see also Baddeley, 2003), but VSWM to be implied in the construction of the situational model from spatial texts.

Once a spatial model is activated from a spatial description, and is supposed to represent the spatial layout of landmarks, does it need VSWM memory for reactivation? So far, to the best of our knowledge, no studies have directly addressed whether VSWM is implied in retrieval of spatial mental models. Suggestions on this issue derive from studies of the production of route directions and spatial descriptions. In Levelt’s theory (Levelt, 1989) the very first phase of speech

generation (*conceptualising*) consists of: conceiving a communicative goal, selecting the relevant information to be expressed for realising this goal, and organising this information. All these processes have a preverbal message as output, and operate on procedural and declarative knowledge stored in working and long-term memory. Extending this model to the production of route directions, Denis (1996) claimed that these operations consisted of the activation of an internal representation of the spatial environment to be described. The internal representation will probably include procedural information and environmental visual-spatial features (Denis, Daniel, Fontaine, & Pazzaglia, 2001). If this were the case, we would expect VSWM to be involved in activation of the internal spatial representation and, consequently, VSWM to have a role not only in spatial language encoding, but also in spatial language production.

Investigating the role of VSWM and VWM in the retrieval of an internal representation derived from a spatial description is our main objective. To this end, in two experiments, we asked participants to listen to spatial and nonspatial texts, performing spatial (Experiment 1) or verbal (Experiment 2) concurrent tasks during encoding or retrieval. In Experiment 1, we aimed to check the hypothesis that VSWM is more involved in encoding and retrieval of the spatial text than of the nonspatial text. Support would come from a selective interference effect of the spatial concurrent task, limited to encoding and retrieval of the spatial text. Experiment 2 was designed to confirm the results of Experiment 1 by demonstrating that these were actually due to the spatial nature of the concurrent task. Moreover, we expected that switching the concurrent task from spatial to verbal would produce a difference in the pattern of findings compared to those of Experiment 1. In the last part of the paper, the data from Experiments 1 and 2 undergo general re-analysis, examining the relative impact of the verbal and visual-spatial components of WM during encoding and retrieval of the spatial text.

Experiment 1

Experiment 1 aimed to test the hypothesis that the final product of spatial text processing is a spatial mental model whose construction and reactivation require involvement of VSWM. To this end, we used a dual-task paradigm. The main task required comprehension of two different texts; the concurrent task was a ST, which consisted in sequential tapping of the same four keys on a computer keyboard hidden from the participant’s view. This task was intended to compete with maintenance of information in VSWM (Logie, 1995). Its effect on the performance of different spatial tasks has been demonstrated in several experiments (Deyzac, Logie, & Denis, 2006; Garden, Cornoldi, & Logie, 2002; Logie & Salway, 1990; Quinn & Ralston, 1986; Smyth & Pendleton, 1989). It was expected that performance of the ST

task would interfere with the construction and reactivation of the spatial mental model derived from processing the spatial text. As a consequence, we expected the concurrent spatial task to have a stronger impairment effect on the encoding and retrieval of the spatial text, compared to the nonspatial text.

Method

Participants

A total of 54 (3 males, 51 females) undergraduates from the Faculty of Psychology of the University of Padua participated. Mean age was 23 years. They were randomly assigned to three groups: with dual-task during encoding, 17 participants (1 male, 16 females); dual-task during retrieval, 19 participants (1 male, 18 females); control group, no dual-task, 18 participants (1 male, 17 females).

Materials

Texts

Two different texts were used, one spatial, one nonspatial. The texts were those used in De Beni et al. (2005), with minimal formal revisions: English translations of parts of the Italian originals are given in Appendix 1. The spatial text described a farm from the perspective of a person walking round the property. A total of 10 landmarks were mentioned, and for each the position was clearly specified from the navigator's perspective. The nonspatial text was a procedural text listing the sequence of operations to be followed for production of various kinds of wine. Although it described a physical procedure, it contained many abstract words such as "temperature" and "fermentation", which should make it harder for the reader to have a visual representation of its content.

The two texts were the same length: 269 words for the spatial text, 260 for the nonspatial text. They were devised to be similar as regards the number of units of information (i.e. 10). In the spatial text these comprised spatial landmarks and their positions; in the nonspatial text they were pieces of information essential to understanding the texts and the specifications in them. The 10 units of information in the nonspatial text were decided on by asking three independent judges to read the text and to list the most important pieces of information. Those chosen by at least two of the judges were used for scoring. Text difficulty was tested in a pilot experiment during which undergraduates read them and subsequently performed free-recall and sentence-verification tasks on the contents. The number of pieces of information reported in the free-recall test and of correct responses for the sentence-verification task was the same

for the two texts, confirming comparable level of difficulty.

Sentence-verification task

The sentence-verification task consisted of 30 sentences for each text, half true and half false, all having the same length. The sentences, similar to those used in Taylor and Tversky (1992), were not taken verbatim from the texts, nor were they just paraphrased versions of the original sentences: instead, they contained information not explicitly stated in the texts. Hence, in verifying their truthfulness responders were supposed to make inferences based on information provided in the text. Examples of sentences are: "With respect to the well, the barn is in the farthest corner" (True), "During fermentation the new wine is stored at sub-zero temperatures" (False).

Concurrent spatial task

The device for the ST task consisted of four keys corresponding to numbers 1, 3, 9, and 7 on a digital keyboard. The four keys were displayed in the four corners of a virtual square whose side was 50 mm long. The keys had standard shape and size (a square with 12 mm sides) and the distance between them was 26 mm. A special box covered the keyboard, hidden from the participant's view, and left an opening for the hand to tap.

Design

The design was mixed, with the concurrent task (encoding vs. retrieval vs. control) as a between-participants factor, and the text, spatial versus nonspatial, as a within-participant factor.

Procedure

Participants were tested individually for about 50 min. They were told that the experiment required them to listen to and memorise two texts. In the two interference groups, they were also told about the concurrent task.

The participants listened to each text twice. Text presentation was counterbalanced across participants. Just after the second listening to each text, they performed, in order, free-recall and sentence-verification tasks on the content of the text. The free-recall task required participants to give a verbal report of all text information they could recall. Verbal recalls were tape-recorded. In the verification task the experimenter read the 30 sentences in random order and for each the participant gave an immediate verbal response of "true" or "false". Responses were recorded on paper by the experimenter.

The ST consisted in tapping sequentially the four keyboard keys described above with the finger of the dominant hand in a counter-clockwise order. No particular rate was imposed. The computer program recorded the sequences of digits tapped throughout performance of the entire task, and the experimenter recorded the time from the beginning to the end of the task. When the participant made an error in the sequential tapping for more than two consecutive sequences, the experimenter repositioned the participant's finger on the "1" key. The two interference groups practised the ST for 30 s before starting the experiment.

The participants assigned to the encoding condition performed the ST while listening to the texts, whereas those in the retrieval group performed the ST concurrently with the free-recall and the sentence-verification tasks. The control group had no concurrent task.

Scoring and data analysis

The free-recall protocols were scored by calculating the number of units correctly recalled. Two independent judges, using the predefined scheme of 10 information units described in the material section, scored the protocols of 20 participants. Depending on the participant's answer, points between 0 and 2 were assigned. For the spatial text, one score was assigned when a participant named a landmark and gave an approximate indication of its position ("After the barn, I will find the poultry pen"); two scores when a participant named a landmark and gave a precise indication of its position ("After the barn, I will see the poultry pen on my right"). For the nonspatial text, one score was assigned when a unit of information was recalled but with no further specification ("To produce the red wine the grapes are left to ferment"); two scores when the unit of information was recalled and specified in full ("To produce the red wine the grapes are left to ferment at 15–18°C"). Given that the (Pearson's) correlation between the scores of the two judges was 0.83 and 0.93 for spatial and nonspatial texts, respectively, $p < 0.001$, the remaining protocols were scored by one of the two judges (the experimenter), whose scores have been analysed. The sentence-verification task was scored by calculating the number of sentences correctly verified.

The concurrent spatial task performances were analysed using t tests comparing errors (percentage of sequences not correctly tapped out of the total of tapped sequences) and time (mean time in seconds for each sequence) when the spatial task were performed concurrently with the encoding and retrieval of the spatial and nonspatial texts.

Scores of the free-recall and sentence-verification tasks were analysed using separate 2×3 mixed ANOVAs having text as within-participant factor and concurrent task as between-participants factor. Post hoc comparisons between means were carried out using Tukey's tests. In order to evaluate the effect size of the concurrent task

during text encoding or retrieval, we calculated Cohen's (1988) d indices for each of the comparisons between the baseline and the two interference groups.

Results

Concurrent spatial task

We calculated the mean percentage of errors that occurred in the concurrent ST task when it was performed during encoding (listening to the text) and retrieval (free-recall and sentence-verification tasks), for the two texts.

Of participants with mean percentage of errors $> 10\%$ (range 10.18–42.34) there were four in the encoding condition (two with nonspatial text, two with spatial text), seven in the retrieval-free-recall (two with nonspatial text, five with spatial text), and five in the retrieval-sentence-verification (two with nonspatial text, three with spatial text). It thus seems that the concurrent spatial task was difficult and not completely automatic, at least for some of the participants. We then verified whether processing the spatial and nonspatial texts had a different impact on errors performed on the concurrent tapping task. Listening to the spatial and nonspatial texts had the same impact on the spatial concurrent task, $t(17) < 1$. Similarly, no differences due to texts were found when the tapping task took place during the free-recall and the sentence-verification tasks: $t(17) = 1.48$, $p = 0.15$; $t(17) < 1$, respectively. We calculated the mean times in seconds for each sequence of the tapping task when it occurred during the encoding, free-recall and verification tasks of the spatial and nonspatial texts. Comparisons between means did not result in significant differences: encoding $t(16) < 1$; free-recall $t(13) = 1.70$, $p = 0.11$; verification $t(13) < 1$.

Text memory and comprehension

Free-recall

The mean overall scores assigned to information units correctly recalled for the spatial and nonspatial texts for the three groups (concurrent task during encoding, retrieval, and control) and corresponding standard deviations are presented in Table 1.

The ANOVA results showed a significant effect of text, $F(1,51) = 92.47$, $\eta^2 = 0.64$, $p < 0.001$, owing to a better performance on the nonspatial text compared with the spatial text. The effect was due to the specific impact of the spatial interference (exemplified by the interaction described below), since in the control condition the spatial and nonspatial texts resulted in identical performances. The main effect of the concurrent task was also significant, $F(2,51) = 23.11$, $\eta^2 = 0.47$, $p < 0.001$. The three means differed from each other significantly (Tukey, $c.d. = 3.46$, $p < 0.05$), with the best performance in the control condition, followed by that

Table 1 Experiment 1: means and standard deviations overall scores assigned to information units correctly recalled for the spatial and nonspatial texts for the concurrent spatial task during encoding, retrieval, and for control groups

		Spatial text	Nonspatial text	Total concurrent task
Control condition	Mean	12.50	13.61	13.05
	SD	4.19	3.86	4.02
Encoding	Mean	1.11	8.29	4.71
	SD	2.14	5.06	3.60
Retrieval	Mean	6.31	12.26	9.29
	SD	4.37	4.09	9.28
Total text	Mean	6.74	11.46	
	SD	5.19	4.82	

with interference during retrieval, and lastly performance with interference during encoding. Moreover, the results showed the expected text×concurrent task interaction, $F(2,51) = 13.93$, $\eta^2 = 0.35$, $p < 0.001$.

The comparison between pairs of means (Tukey, *c.d.* = 1.72, $p < 0.05$) showed that the ST, during both encoding and retrieval, impaired the performance of the spatial text more than that of the nonspatial text: the spatial text was significantly less well recalled than the nonspatial text in the two concurrent task conditions, whilst no difference emerged between the two texts in the control condition.

The comparison between performances for each text in the three conditions showed that the concurrent spatial task affected the two texts differently in encoding and retrieval. For the spatial text, the control and the two interference conditions were found to differ from each other, the best performance being in the control, followed by interference during retrieval and interference during encoding. For the nonspatial text, control and retrieval did not differ and both were significantly better than encoding.

Cohen's (1988) *d* indices were calculated relative to the comparisons between the baselines (controls) and the concurrent task conditions for both spatial and nonspatial texts.

In both comparisons, baseline versus encoding and baseline versus retrieval, the effects on the spatial text were larger than those on the nonspatial text: the ST during the encoding of the spatial text caused a fall in performance of 1.71 SD respect to the control, whereas the fall was 1.02 SD for the nonspatial text. Similarly the ST during retrieval impaired the performance of the spatial text by 1.18 SD whereas the decrement was 0.34 SD for the nonspatial text.

Verification task

Table 2 shows the average number of sentences correctly verified for the spatial and nonspatial texts as a function of concurrent task.

The main effects of text and group were significant. The main effect of text, $F(1,51) = 25.11$, $\eta^2 = 0.33$, $p < 0.001$, showed that the performance for the non-

Table 2 Experiment 1: means and standard deviations (percentages in brackets) of the number of sentences correctly verified for the spatial and nonspatial texts for the concurrent spatial task during encoding, retrieval, and for control groups

		Spatial text	Nonspatial text	Total concurrent task
Control condition	Mean	25.00 (83%)	25.33(84%)	25.17 (84%)
	SD	4.19	3.43	3.69
Encoding	Mean	17.47 (58%)	20.88(70%)	19.19 (64%)
	SD	4.91	5.06	4.41
Retrieval	Mean	21.31 (71%)	24.63(82%)	22.97 (77%)
	SD	4.48	2.61	3.30
Total text	Mean	21.33 (71%)	23.68(79%)	
	SD	5.39	4.19	

spatial text was higher than that the spatial text. The main effect of the concurrent task $F(2,51) = 11.06$, $\eta^2 = 0.303$, $p < 0.001$ showed that the performances in control and retrieval did not differ and that both were significantly better than in encoding (Tukey, *c.d.* = 3.62, $p < 0.05$). The results revealed also the expected text×concurrent task interaction $F(2,51) = 4.63$, $\eta^2 = 0.15$, $p = 0.014$.

The comparison between pairs of means showed that, while in the control condition there was no difference between numbers of sentences correctly verified for either spatial or nonspatial texts, their number was significantly lower for the spatial text when there was a concurrent spatial task during encoding and during retrieval (Tukey, *c.d.* = 1.64, $p < 0.05$).

Comparison between performances of each text in the three conditions showed two distinct patterns of results for the spatial and nonspatial texts, similar to those found in the free-recall. For the spatial text, the performance in the three conditions differed, the best performance being in the control condition and the worst with ST during encoding, whereas in the nonspatial text control and interference during retrieval they did not differ, and both were significantly better than with interference during encoding.

Similarly to the free-recall task, comparison between Cohen's (1988) *d* indices showed a higher effect size on the spatial texts than the nonspatial texts. In both comparisons, baseline versus encoding and baseline versus retrieval, the effects on the spatial text exceeded those on the nonspatial text: the ST during encoding of the spatial text causes a decrement in performance of 1.28 SD, whereas the decrement was of 0.93 SD for the nonspatial text. Similarly, the ST during retrieval impaired the performance for the spatial text by 0.79 SD, whereas the decrement was 0.23 SD for the nonspatial text.

Experiment 2

To clarify the extent to which the results of Experiment 1 were due to the involvement of VSWM in a spatial text processing, we performed Experiment 2 to check whether a different concurrent task, verbal instead of spatial,

had the same or different effect on the two texts. We followed the same procedure as in Experiment 1, except for the concurrent task, which was an AS (i.e. the repetition of five syllables), whose effect in competing with maintenance of phonological traces in VWM has been clearly demonstrated (Baddeley, 1990; Baddeley, Lewis, & Vallar, 1984; Levy, 1971, 1975). We expected the change in concurrent task to alter the results. Since comprehension of both texts required processing of verbal material, we expected that, in contrast with the concurrent spatial task, the verbal task would have a similar, nonspecific, interference effect on both nonspatial and spatial texts (De Beni et al., 2005). More precisely, we expected that the AS performed during encoding would impair both texts to the same extent, given the widely demonstrated role of VWM in text processing (Gathercole & Baddeley, 1993; McCarthy & Warrington, 1987; Vallar & Baddeley, 1984, 1987). With the AS during retrieval we again expected a different pattern of results. Whereas in Experiment 1 the ST had a detrimental effect on retrieval of the spatial text, we expected the AS to have minimal effect in Experiment 2. We hypothesised that, once a spatial mental model is constructed, it should not need a verbal articulatory component to reactivate it: text contents were processed and organised during encoding, in a format independent from surface text format (van Dijk & Kintsch, 1983), and retrieval should imply the mechanism of rehearsal on the phonological loop to only minimum extent.

Method

Participants

A total of 50 (11 males, 39 females) undergraduates from the Faculty of Psychology of the University of Padua participated. Mean age was 23 years. They were randomly assigned to three groups having: dual-task during encoding, 16 participants (5 male, 11 females), dual-task during retrieval, 16 participants (5 male, 11 females), control group, no dual-task, 18 participants (1 male, 17 females).

Materials

Texts and sentences: as for Experiment 1.

Concurrent verbal task

Repetition of the series ba-be-bi-bo-bu was chosen as an AS task.

Design

Design as for Experiment 1.

Procedure

As for Experiment 1, except for two changes:

1. This regarded the experimental manipulation and consisted of the use of the AS as a concurrent verbal task. Participants had to train the secondary task simply by repeating the sequence of syllables twice. In the encoding condition they listened to the texts and concurrently continued to repeat ba-be-bi-bo-bu. After presentation of each text, they stopped the concurrent verbal task and had to perform the free-recall and sentence-verification tasks. The group with the concurrent verbal task during retrieval was asked to perform the AS during the performance of both free-recall and sentence-verification tasks.
2. This regarded the way responses were given in the two memory tasks. In the free-recall task, participants had to write down all the relevant information they remembered (instead of reporting it verbally), and in the verification task they had to verify the true/false sentences, presented one at a time in random order on paper, by ticking T or F on a response form (instead of stating true or false verbally for each sentence). Participants had 10 s to read each sentence. The changes in response mode were considered necessary to avoid competition between verbal responses and execution of the AS task.

Scoring procedures and data analysis

As for Experiment 1, and following the same criteria, two independent judges scored 20 free-recall protocols. Since the correlation between the scores of the two judges was 0.82 and 0.96 for spatial and nonspatial texts, respectively, $p < 0.001$ for both, one of the judges (the experimenter) continued to score the protocols and the scores were analysed.

Results

Concurrent verbal task

The participants made no errors during performance of the concurrent task, so only times were analysed. We calculated the mean times in seconds for syllable sequence when the AS was performed during encoding and retrieval of the two texts. Comparisons between performance on the concurrent verbal task during processing of the spatial and nonspatial texts did not result in significant differences: encoding $t(15) < 1$, free-recall $t(15) < 1$; verification task $t(15) = 1.11$, $p = 0.28$.

Text memory and comprehension

Free-recall

Table 3 Experiment 2: means and standard deviations overall scores assigned to information units correctly recalled for the spatial and nonspatial texts for the concurrent verbal task during encoding, retrieval, and for control groups

		Spatial text	Nonspatial text	Total concurrent task
Control condition	Mean	13.22	12.67	12.94
	SD	4.27	4.52	4.39
Encoding	Mean	6.31	6.43	6.37
	SD	5.62	5.36	5.49
Retrieval	Mean	10.87	8.06	9.46
	SD	6.93	5.44	6.18
Total text	Mean	10.26	9.20	
	SD	6.26	5.68	

We first compared the performance of control groups in Experiments 1 and 2 to exclude the possibility that the different response modes (verbal in Experiment 1, written in Experiment 2) affected comprehension scores. The mean number of units correctly recalled for the spatial text was 12.50 (SD=4.19) in Experiment 1, and 13.22 (SD=4.28) in Experiment 2. The difference was not significant: $t(37) < 1$. The mean number of units correctly recalled for the nonspatial text was 13.61 (SD=3.87) in Experiment 1, and 12.67 (SD=4.52) in Experiment 2. The difference was not significant: $t(37) < 1$.

The mean overall number of information units correctly recalled for the spatial and nonspatial texts for the groups with the concurrent task in encoding, retrieval, or control, and relative standard deviations are shown in Table 3.

An ANOVA showed a significant main effect of concurrent task, $F(2,47)=9.28$, $\eta^2=0.283$, $p<0.001$. As expected, recall with interference during encoding was more impaired than in the control condition (Tukey, $c.d.=4.41$, $p<0.05$), and no difference was found between performances in the control condition and with interference during retrieval. Moreover, as expected, no significant effects were found, of text ($F(1,47)=1.56$, $\eta^2=0.032$, $p=0.22$) and of the interaction text $\times$ concurrent task ($F(1,47)=1.02$, $\eta^2=0.042$, $p=0.36$).

Cohen's (1988) d indices confirmed the results of the statistical analyses. Comparisons between control and encoding for the nonspatial and spatial texts had very similar effect sizes suggesting an equal load of the concurrent verbal task on the two texts (1.07 and 1.15, for nonspatial and spatial texts, respectively).

Verification task

As regards control group performance, the mean number of sentences correctly verified for the spatial text was 25.00 (SD=4.19) in Experiment 1, and 23.33 (SD=3.07) in Experiment 2. The difference was not significant: $t(37)=1.09$, $p=0.28$. Mean number of units correctly recalled for the nonspatial text was 25.33 (SD=2.43) in Experiment 1, and 24.17 (SD=2.99) in Experiment 2. The difference was not significant: $t(37)=1.36$, $p=0.18$.

The mean overall number of sentences correctly verified for the spatial and nonspatial texts for the three groups and relative standard deviations are shown in Table 4.

The results showed a marginally significant main effect of the concurrent task, $F(2,47)=3.13$, $\eta^2=0.117$, $p=0.053$. A comparison between pairs of means (Tukey, $c.d.=2.05$, $p<0.05$) showed that performance with interference during encoding was significantly worse than that in control (this difference was marginally significant) and with interference during retrieval. As expected, control and retrieval did not differ. A significant main effect of text was found, $F(1,47)=7.23$, $\eta^2=0.133$, $p=0.01$: the recall of the nonspatial text was higher than that of spatial text. In accordance to our expectations, the interaction text $\times$ concurrent task was not significant ($F(2,47)=0.595$, $\eta^2=0.025$, $p=0.556$).

Cohen's (1988) d indices revealed quite a large effect of the concurrent verbal task on the encoding of the spatial text (0.75), whereas the effect was smaller on the encoding of the nonspatial text (0.39). The effect of the concurrent verbal task during retrieval was low for both spatial (0.05) and nonspatial texts (0.21).

Comparison between the two experiments

Analysis of the data collected in the two experiments on free-recall and on verification task allowed comparison of the impact of the spatial and verbal concurrent tasks ("kind of concurrent task") on the encoding and retrieval of information from the two texts. Below, we do not report statistically significant effects already given in the two previous Results sections, and have instead focused on effects due to the new factor kind of concurrent task.

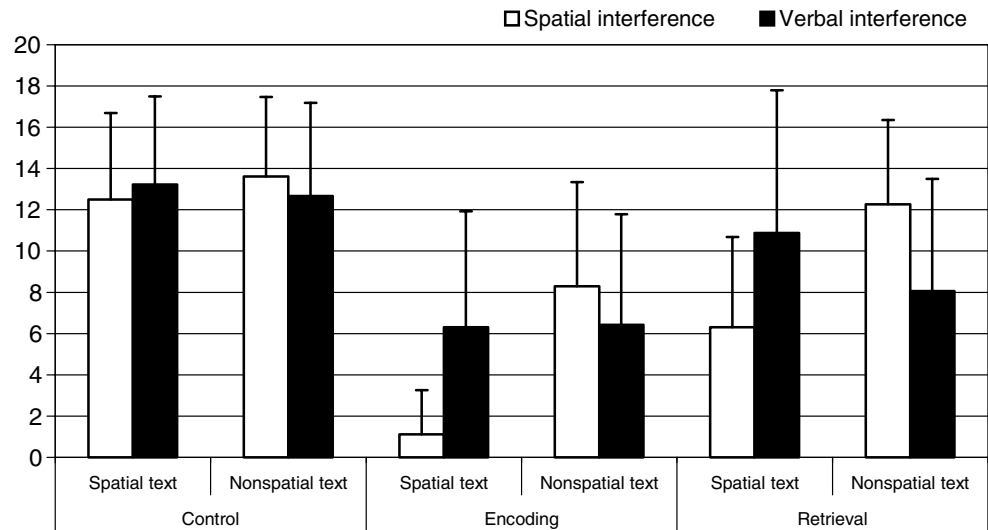
Free-recall

The mean number of overall information units correctly recalled for the two texts as a function of the two

Table 4 Experiment 2: means and standard deviations (percentages in brackets) of the number of sentences correctly verified for the spatial and nonspatial texts for the concurrent verbal task during encoding, retrieval, and for control groups

		Spatial text	Nonspatial text	Total concurrent task
Control condition	Mean	23.33 (78%)	24.16 (81%)	23.75 (79%)
	SD	3.07	2.99	3.03
Encoding	Mean	20.56 (69%)	23.00 (77%)	21.78 (73%)
	SD	3.86	3.06	3.46
Retrieval	Mean	23.12 (77%)	24.75 (82%)	23.93 (80%)
	SD	4.76	2.72	3.74
Total text	Mean	22.38 (75%)	23.98 (80%)	
	SD	4.04	2.96	

Fig. 1 Means and standard deviations of the overall number of information units recalled for the two texts as a function of the verbal and spatial concurrent tasks in encoding and retrieval conditions



concurrent tasks during encoding retrieval and in the control condition are presented in Fig. 1.

The results showed significant effects of text (spatial vs. nonspatial) $\times$ kind of concurrent task (spatial vs. verbal), $F(1,98) = 35.62$, $\eta^2 = 0.27$, $p < 0.001$ and of text (spatial vs. nonspatial) $\times$ concurrent task (encoding vs. retrieval vs. control) $\times$ kind of concurrent task (spatial vs. verbal), $F(2,98) = 4.92$, $\eta^2 = 0.09$, $p < 0.01$.

The interaction text (spatial vs. nonspatial) $\times$ kind of concurrent task (spatial vs. verbal) showed that with the spatial concurrent task, performance was lower in the spatial text than the nonspatial text, but no difference was found between texts with verbal interference (Tukey, *c.d.* = 1.40, $p < 0.05$; spatial interference: spatial $M = 6.64$, $SD = 3.57$ and nonspatial $M = 11.38$, $SD = 4.33$; verbal interference: spatial $M = 10.13$, $SD = 5.61$ and nonspatial $M = 9.05$, $SD = 5.11$). The third-order interaction showed that in the control condition performance was similar with the spatial and the nonspatial text in both experiments. However, with a concurrent task during encoding, performance on the spatial text was worse with the ST than with the AS, while the converse was true for the nonspatial text. With a concurrent task during retrieval, performance on the spatial text was worse with the spatial interference task than with the verbal task, while the converse was true for the nonspatial text (Tukey, *c.d.* = 1.71, $p < 0.05$).

The effect sizes of the concurrent task (encoding vs. retrieval vs. control) conditions in the two experiments confirm the results of the overall analysis. In comparison between control and encoding, the AS resulted in similar effect sizes for the spatial and nonspatial texts (1.15 and 1.07, respectively). In contrast, the ST had a larger effect on the spatial text than the nonspatial text (1.71 and 1.02, respectively). With interference during retrieval, the AS produced a larger effect on the nonspatial text (0.84), double that on the spatial text (0.41). In contrast, the ST had a large effect on the spatial text (1.18), three times that for the nonspatial text (0.34).

Verification task

The mean overall number of sentences correctly verified from the two texts as a function of the two concurrent tasks during encoding, retrieval and in the control condition are presented in Fig. 2.

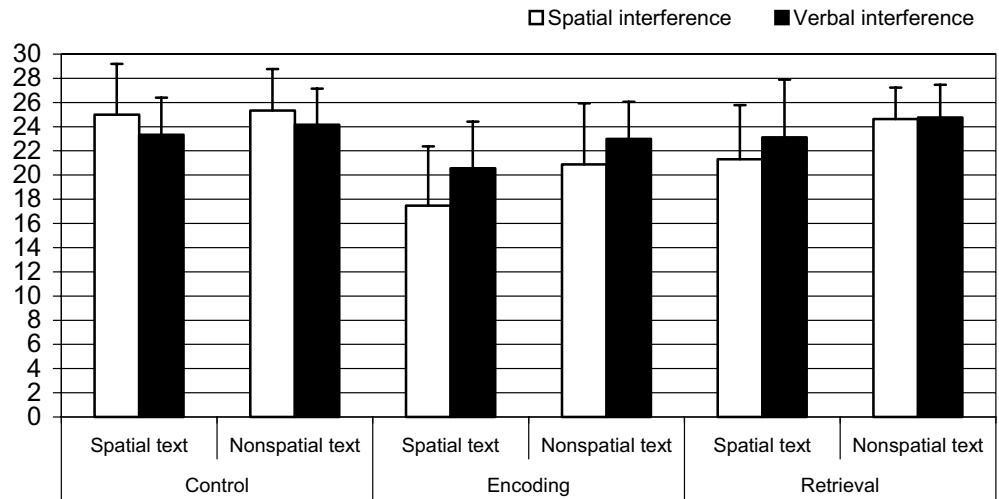
The results showed a significant effect of text (spatial vs. nonspatial) $\times$ concurrent task (encoding vs. retrieval vs. control), $F(2,98) = 3.60$, $\eta^2 = 0.07$, $p < 0.05$ and of concurrent task (encoding vs. retrieval vs. control) $\times$ kind of concurrent task (spatial vs. verbal) was also significant, $F(2,98) = 3.19$, $\eta^2 = 0.06$, $p < 0.05$.

The interaction text (spatial vs. nonspatial) $\times$ concurrent task (encoding vs. retrieval vs. control) showed that the performance of the two texts did not differ in the control condition (Tukey, *c.d.* = 1.34, $p < 0.05$; spatial $M = 24.16$, $SD = 3.63$; nonspatial $M = 24.74$, $SD = 3.21$) but that of the spatial text was worse both in encoding and retrieval (encoding: spatial $M = 19.01$, $SD = 4.38$; nonspatial $M = 21.94$, $SD = 4.06$; retrieval: spatial $M = 22.21$, $SD = 4.62$; nonspatial $M = 24.69$, $SD = 2.66$).

The interaction concurrent task (encoding vs. retrieval vs. control) $\times$ kind of concurrent task (spatial vs. verbal) showed (Tukey, *c.d.* = 2.29, $p < 0.05$) that for encoding only, the spatial concurrent task ($M = 19.17$, $SD = 4.98$) impaired the texts more than the verbal ($M = 21.78$, $SD = 3.46$), whereas no differences were found in control and retrieval (control: spatial interference $M = 25.16$, $SD = 3.81$, verbal interference $M = 23.74$, $SD = 3.03$; retrieval: spatial interference $M = 22.97$, $SD = 3.54$, verbal interference $M = 23.93$, $SD = 3.74$).

Considering the effect size, with interference during encoding we had a high, and similar, effect of the spatial tapping on both texts (0.93 and 1.28, for nonspatial and spatial texts, respectively). The effect on the spatial text was larger, but the same pattern was obtained with the concurrent verbal task, which had an effect size 0.39 for the nonspatial text and 0.75 for the spatial text. It seems

Fig. 2 Means and standard deviations of the overall number of sentences correctly verified for the two texts as a function of the verbal and spatial concurrent tasks in encoding and retrieval conditions



that the impact of both concurrent tasks was higher on the spatial compared to the nonspatial text. With the concurrent tasks during retrieval, the data goes towards the direction of a specific interference effect of the spatial concurrent task on the spatial text. In fact whereas the verbal concurrent task had very low effects on both the two texts (0.21 and 0.05, for nonspatial and spatial texts, respectively), the spatial concurrent task had a high effect (0.79) on the spatial text, much larger than that on the nonspatial text (0.23).

General discussion

In our study, we have investigated the involvement of VSWM and VWM during encoding and retrieval of two texts; one spatial, one nonspatial. Research carried out to date on mental models has looked at the role of VWM and VSWM in the construction of mental models derived from illustrated texts (Gyselinck, Cornoldi, Dubois, De Beni, & Ehrlich, 2002; Gyselinck, Ehrlich, Cornoldi, De Beni, & Dubois, 2000; Kruley, Sciana, & Glenberg, 1994), in spatial descriptions (De Beni et al., 2005; Pazzaglia & Cornoldi, 1999) and in a sentence-picture verification task (Noordzij et al., 2004). These studies found that VSWM was specifically involved in the processing of illustrated texts, spatial sentences (Brooks, 1967) and spatial descriptions (De Beni et al., 2005). Other studies, which analysed the features of the spatial mental models activated during production of route directions (e.g. Daniel, Tom, Manghi, & Denis, 2003; Daniel & Denis, 2004), suggested that the various components of WM constitute a cognitive support during retrieval of the mental representations that enabled their production.

However, to the best of our knowledge, the role of VSWM and VWM during retrieval of spatial descriptions had not yet been directly investigated. The present study has brought fresh evidence as regards the specific involvement of VSWM in the retrieval of spatial infor-

mation. In Experiment 1, we examined the effects of a concurrent ST, assumed to interfere with the consolidation of a visual trace in the VSWM, on the encoding and retrieval of spatial and nonspatial texts. We expected the ST to impair the encoding of information for the spatial text and also its retrieval, more than for the nonspatial text. The results supported our hypothesis that the performance of ST and the construction/reactivation of a spatial model share the same resources of VSWM, since the spatial text was recalled more poorly and verified less accurately than the nonspatial text only in the two concurrent task conditions, whilst it was equally recalled and verified in the control.

The role of VSWM in the retrieval of the spatial text looked clear-cut, since the ST interfered with comprehension and memorisation of the spatial text, but not the nonspatial text. What about the effect of the ST in encoding? Our results pointed to a detrimental effect of the ST on the performance of both texts, even if more dramatic for the spatial text. These findings suggest that the ST has a nonspecific detrimental effect on the encoding of both texts, along with a specific impact on the encoding of the spatial text only. The nonspecific effect may be due to attentional resources shared by performing the ST and text processing. However, to our knowledge, similar effects of a ST have not been reported in the literature. An alternative explanation is that the encoding of the nonspatial text also benefited from visual-spatial strategies such as the construction of mental models (e.g. spatial linear ordering) that were disrupted by the spatial secondary task. Nevertheless, explanation is still needed of why De Beni et al., who used similar materials, but a different ST, failed to find any impairment effect of the ST on the memorisation of the nonspatial text. Further research should clear up this point.

The results of Experiment 2 help in understanding the role of VWM in the encoding and retrieval of information from spatial and nonspatial texts. The outcomes of the free-recall task are compatible with the idea that a verbal interference task has the same impairment effect

on the encoding of both spatial and nonspatial texts (see also De Beni et al., 2005). The interaction between text and concurrent task was, as expected, not significant, suggesting that the two texts are affected to the same extent by the concurrent verbal task. A comparison between Cohen's (1988) *d* indices confirms the statistical analyses. The AS had very similar effects on both nonspatial and spatial texts. Interestingly, we found that the AS does not hamper performance when it occurs during text retrieval. This is in line with multi-level models of text processing (Kintsch & van Dijk, 1978) and adds fresh support to the idea that a mental model, whether spatial or not, needs the articulatory loop for its construction, but not for reactivation.

As for the sentence-verification task, the difference between the control and concurrent task conditions during encoding was only marginally significant. It seems that the free-recall was more sensitive than the sentence-verification to the detrimental effects of the AS during text encoding. We do not believe this to be due to features inherent in the sentences used in the sentence-verification task. In fact, as a true/false task, the difficulty was adequate (about 80% of correct responses in the control conditions) and the performance produced the expected effects in Experiment 1. It is possible that the participants, owing to the prompt given by the sentences, were able to re-analyse information just processed and so reconstruct the mental model when sentences were presented in the sentence-verification task. A similar process, referred to as *backward updating*, was described by de Vega (1995) in a study that investigated a reader's ability to track a protagonist's position during continuous reading. Overall, our results showed that VSWM and VWM are differently involved in the encoding and retrieval of spatial and nonspatial texts. The role of VSWM on retrieval supports the idea that this process starts from an internal representation with spatial features (see Denis, 1996; Denis & Cocude, 1997; Denis & Denhière, 1990 for spatial language production) mediated by the visual-spatial component of WM. We used a split design, with the ST in Experiment 1 and AS in Experiment 2, because a global design would have implied dealing with at least three main effects and relative interactions. However, the last part of the study, in which data from the two experiments are globally analysed, allowed us to overcome limitations of the split design and give a more direct answer to the question of the extent to which VSWM and VWM are involved in spatial and nonspatial text processing. Comparisons of the involvement of the two components in spatial text encoding and retrieval revealed that VSWM had a prominent role. These results confirmed those obtained by De Beni et al. (2005) and Gyselinck et al. (2000, 2002) for spatial text processing, and extended the role of VSWM to retrieval.

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Appendix 1

Spatial text

Imagine yourself standing in a meadow dotted with flowers in front of the tall boundary walls of a holiday farm called "Tranquillity" stretching over a rectangular area.

On your right, on the corner of the holiday farm, there is the entrance gate...

...Once you have reached the restaurant, turn left and, leaving the restaurant behind, you will soon pass a little bridge crossing a small lake...

...When you get there, turn left leaving the barn behind you and continue on your way for a while. You will see the poultry pen on your right...

...Turn left and continue walking for a long way along the brick wall; keeping the vineyard on the left. You will then get back to the gate.

Nonspatial text

The grape harvest is performed at different times of the year depending on the quality of the wine: sparkling wine during the first week of September, and non-sparkling wine during the second week of September...

...Non-sparkling wines are obtained from fully matured grapes and have a high alcohol level and low acidity...

...There are two types of vinification process, i.e. two different ways to make wine, for red and white wine.

...Before bottling, crystallisation takes place by bringing the wine to sub-zero temperatures, about -5°C . This procedure lasts 2 days and allows the excess tartar to deposit so it can be eliminated later.

To produce white wines, the grapes are crushed and the skins immediately discarded: the must obtained is put into casks.

Fermentation occurs at $15-18^{\circ}\text{C}$, then crystallisation and bottling takes place.

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